

A Physics-Informed, Regime-Aware Machine Learning Framework for Predicting Adsorption Removal Efficiency

Abstract

Adsorption-based removal efficiency (RE) is governed by complex nonlinear interactions between solution chemistry, adsorbent properties, and operating conditions. Literature-compiled adsorption datasets exhibit strong heterogeneity, making accurate and physically consistent prediction challenging. This study develops a regime-aware, physics-informed machine learning framework for predicting removal efficiency using gradient boosting methods. By explicitly incorporating pH- and concentration-based regime definitions, robust data cleaning, and physically motivated feature engineering, the proposed framework achieves strong predictive accuracy while maintaining interpretability. Among the evaluated models, a carefully regularized Light Gradient Boosting Machine (LightGBM) achieved the highest performance with a test R^2 of 0.847 and RMSE of 12.05, outperforming XGBoost and other baselines.

1. Introduction

Removal efficiency is a key performance metric in adsorption-driven water and wastewater treatment processes. Accurate prediction of RE enables efficient screening of adsorbents and optimization of operating conditions. However, experimental adsorption data reported in the literature vary widely in terms of adsorbent composition, adsorbate species, solution pH, initial concentration, temperature, and hydrodynamic conditions. This heterogeneity introduces strong nonlinearity and regime-dependent behavior, which limits the effectiveness of traditional regression models.

Recent studies have applied machine learning to adsorption modeling; however, many approaches rely on purely data-driven feature spaces without incorporating physical understanding. This can lead to overfitting, spurious correlations, and limited generalization. The present work addresses these issues by integrating domain knowledge into the model design through regime-aware learning and physics-informed feature engineering.

2. Dataset Description

2.1 Data Source

The dataset consists of experimentally reported adsorption studies collected from peer-reviewed literature. Each record corresponds to a unique adsorption experiment conducted under specific operating conditions. The dataset aggregates studies involving different adsorbents, adsorbates, and experimental protocols, resulting in a highly heterogeneous data distribution.

2.2 Target Variable

The target variable is removal efficiency (RE, %), defined as the percentage of contaminant removed from solution during adsorption.

2.3 Input Variables

Raw input variables include surface area of the adsorbent (m^2/g), adsorption capacity (mg/g), initial contaminant concentration (ppm), contact time (min), adsorbent dose (g/L), stirring speed (RPM), initial solution pH, temperature (K), and adsorbate identity.

3. Data Cleaning and Preprocessing

Literature-reported adsorption data frequently contain inconsistent numeric representations, including decimal commas (e.g., 130,4), uncertainty notation (e.g., 80 ± 2), approximate values (e.g., ~ 50), and multiple reported values within a single cell. A robust numeric parsing function was implemented to standardize all numeric entries by extracting the first valid numeric value and converting it to floating-point format.

Rows containing missing values after cleaning were removed. Removal efficiency values were constrained to the physically meaningful range of 0–100% to eliminate invalid or misreported entries.

4. Physics-Informed Feature Engineering

To embed adsorption physics into the learning process, several derived features were introduced. These features capture known thermodynamic, kinetic, and mass-transfer relationships.

Site density represents the effective number of adsorption sites available relative to pollutant loading, while the capacity-to-loading ratio captures saturation effects. Logarithmic transformation of contact time accounts for non-linear adsorption kinetics. A mixing index combining RPM and contact time reflects mass-transfer enhancement, and thermo-capacity coupling represents temperature-dependent adsorption behavior.

5. Regime Definition Based on pH and Concentration

5.1 pH-Based Classification

Solution pH plays a critical role in adsorption through its influence on surface charge, ionization state, and electrostatic interactions. Based on chemical considerations, adsorption conditions were classified into acidic ($\text{pH} < 7$) and neutral ($\text{pH} \geq 7$) regimes. An alkaline regime was not explicitly defined due to insufficient data coverage at high pH values.

5.2 Concentration-Based Classification

Initial contaminant concentration determines the degree of adsorbent saturation and driving force for mass transfer. To account for this effect, the dataset was divided into low and high concentration regimes based on the median initial concentration (C_{med}). Values below the median were classified as Low C_{med} , while values above the median were classified as High C_{med} .

5.3 Final Regimes

Combining pH and concentration classifications resulted in four regimes: Acidic–Low C_{med} , Acidic–High C_{med} , Neutral–Low C_{med} , and Neutral–High C_{med} . This regime-aware formulation enables the model to account for distinct adsorption behaviors under different chemical environments.

6. Restricted-Domain Modeling

Certain regimes contained very few samples, leading to unstable model training and unreliable generalization. To address this issue, regimes with fewer than 10 samples were excluded from model training and evaluation. This restricted-domain strategy reduces noise while preserving physically meaningful operating conditions.

7. Machine Learning Models

Multiple machine learning models were evaluated, including linear regression, random forest, support vector regression (SVR), XGBoost, and LightGBM. Tree-based gradient boosting methods consistently outperformed linear and kernel-based models due to their ability to capture complex nonlinear interactions.

7.1 XGBoost Benchmark

A regime-aware XGBoost model with sample weighting and residual learning achieved a test R^2 of 0.809 and RMSE of 13.44. This model served as a strong benchmark for comparison.

7.2 LightGBM Final Model

LightGBM employs a leaf-wise tree growth strategy, which can capture localized nonlinear patterns more effectively than level-wise growth when appropriately regularized. By carefully constraining the number of leaves and enforcing minimum samples per leaf, overfitting was avoided.

The final LightGBM model achieved a test R^2 of 0.847 and RMSE of 12.05, representing the best predictive performance among all evaluated models.

8. Model Diagnostics and Interpretability

Residual analysis revealed that prediction errors were approximately symmetrically distributed around zero, indicating unbiased model behavior. Residuals plotted against predicted RE showed no strong heteroscedasticity.

SHAP-based interpretability analysis identified temperature, capacity-to-loading ratio, adsorption capacity, and pH as dominant contributors to removal efficiency, consistent with adsorption thermodynamics and surface chemistry principles.

9. Limitations

The proposed framework is limited by the quality and coverage of literature data. Alkaline adsorption conditions were underrepresented and could not be modeled explicitly. Additionally, experimental heterogeneity imposes an upper bound on achievable predictive accuracy.

10. Conclusions

This study demonstrates that incorporating physical insight and chemical regime definitions into machine learning significantly improves the prediction of adsorption removal efficiency. The regime-aware, physics-informed LightGBM model achieved robust performance while maintaining interpretability, making it suitable for practical screening and analysis of adsorption systems.